

Test 3 – Coding Theory – Solutions

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Question 1

(a) By inspecting the Z channel model shown in Figure 1, we can see that $P_{y_0|x_0} = 1$, $P_{y_0|x_1} = p$, $P_{y_1|x_0} = 0$, and $P_{y_1|x_1} = 1 - p$.

(b) Using probability theory, we can write

$$P_{y_0} = P_{x_0,y_0} + P_{x_1,y_0} = P_{y_0|x_0} \cdot P_{x_0} + P_{y_0|x_1} \cdot P_{x_1} = \frac{1+p}{2}.$$

In the same way, we have

$$P_{y_1} = P_{x_0,y_1} + P_{x_1,y_1} = P_{y_1|x_0} \cdot P_{x_0} + P_{y_1|x_1} \cdot P_{x_1} = \frac{0+1-p}{2} = \frac{1-p}{2}.$$

(c) The channel capacity is given by

$$C = \sum_{j=0}^1 \sum_{i=0}^1 P_{x_i,y_j} \cdot \log_2 \left(\frac{P_{y_j|x_i}}{P_{y_j}} \right) = \frac{1}{2} \cdot \sum_{j=0}^1 \sum_{i=0}^1 P_{y_j|x_i} \cdot \log_2 \left(\frac{P_{y_j|x_i}}{P_{y_j}} \right).$$

We can develop this expression using the previous results:

$$C = \frac{1}{2} \cdot \left(\log_2 \left(\frac{2}{1+p} \right) + p \cdot \log_2 \left(\frac{2p}{1+p} \right) + 0 \cdot \log_2(0) + (1-p) \cdot \log_2 \left(\frac{2(1-p)}{1-p} \right) \right),$$

which yields

$$C = \frac{1}{2} \cdot (1 - \log_2(1+p) + p + p \cdot \log_2(p) - p \cdot \log_2(1+p) + 1 - p),$$

and finally

$$C = 1 + \frac{1}{2} \cdot (p \cdot \log_2(p) - (1+p) \cdot \log_2(1+p)) \text{ Shannon/channel use.}$$

(d) For $p = 0$, the Z channel capacity is given by

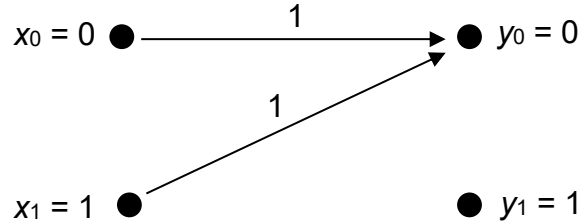
$C = 1 + \frac{1}{2} \cdot (0 \cdot \log_2(0) - (1+0) \cdot \log_2(1+0)) = 1$ Shannon/channel use, which corresponds to a perfect channel that can transmit information with no error.

This result is not surprising since the error probability p is equal to zero.

For $p = 1$, the Z channel capacity is given by

$C = 1 + \frac{1}{2} \cdot (1 \cdot \log_2(1) - (1 + 1) \cdot \log_2(1 + 1)) = 0$ Shannon/channel use, which corresponds to a channel that cannot transmit any information.

This result can be explained by examining the Z channel model obtained for $p = 1$.



The received symbol is always $y_0 = 0$, no matter what the transmitted symbol was. Such channel can obviously transmit no information since the received symbol does not provide us with any information about the transmitted symbol.

Question 2

(a) The equation used to graph the curve entitled “Uncoded BPSK” corresponds to the well-known bit error probability expression of BPSK modulation over an AWGN channel:

$$P_{eb} = \frac{1}{2} \cdot \text{erfc} \left(\sqrt{\frac{E_b}{N_0}} \right).$$

(b) The equation used to graph the curve entitled “(7, 4) Hamming code, soft (ub, 1st term)” corresponds to the first term of the union bound, given by

$$P_{eb} \approx \frac{W_{d_{min}}}{2k} \cdot \text{erfc} \left(\sqrt{d_{min} R_c \frac{E_b}{N_0}} \right).$$

For this Hamming code, we have $d_{min} = 3$, $W_{d_{min}} = 12$, $R_c = \frac{4}{7}$, and $k = 4$. Hence, we can write

$$P_{eb} \approx \frac{12}{2 \times 4} \cdot \text{erfc} \left(\sqrt{\frac{3 \times 4}{7} \frac{E_b}{N_0}} \right) = \frac{3}{2} \cdot \text{erfc} \left(\sqrt{\frac{12}{7} \frac{E_b}{N_0}} \right).$$

(c) This result does indeed make sense. In fact, a channel decoder is always expected to perform significantly better when using soft decisions, i.e. analogue channel samples, rather than hard decisions, i.e. bits generated by a decision block located at the channel output.

The reason is that the decision block takes a threshold decision regarding the value of a transmitted bit by processing the corresponding channel sample as follows: If the channel sample value is above a threshold equal to zero, then the decision block “decides” that the transmitted bit was equal to +1. If this is not the case, the decision block “decides” that the transmitted bit was equal to -1.

Such decision process discards some very useful information that the decoder will no longer be able to exploit. The decoder will perform better if it is able to use the channel samples directly since these samples have not been corrupted by a decision block. Hard-decision decoding should only be employed when the decoding algorithm is, for any reason, unable to process analogue values.

(d) The results shown in Figure 2 are obtained under the implicit assumption that the maximum-likelihood (ML) decoding algorithm is used at the receiver side. Such algorithm provides optimal performance in terms of error correction and should always be used if possible. For some complex channel codes, the size of the message and codeword is so large that the practical implementation of ML decoding is too complex. In such cases, we must use sub-optimal (non-ML) decoding algorithms that offer an error performance which is poorer than that obtained using ML algorithms.

(e) As with any other channel code, the price paid for the error performance improvement due to the use of coding is a reduction in actual data rate or, equivalently, an increase in bandwidth.

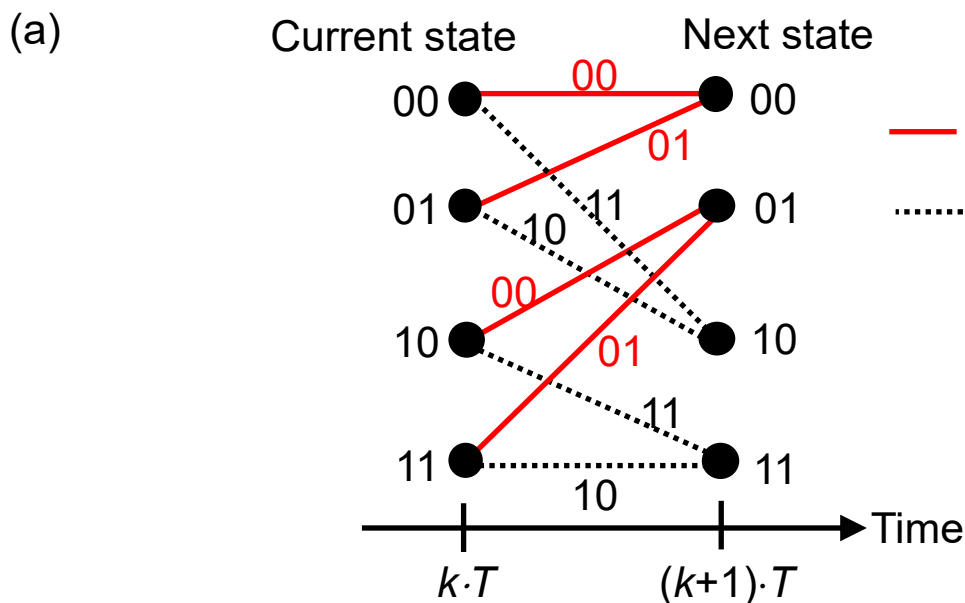
An uncoded system transmits information bits at a rate of D_i^u bits/sec. The transmission rate over the channel is D_c^u bits/sec with $D_c^u = D_i^u$ since the information bits are directly transmitted over this channel. The bandwidth necessary for transmission is B_u and is proportional to D_c^u ($B_u \propto D_c^u$, fundamental principle of digital communications).

When a channel encoder with a coding rate $R_c = \frac{k}{n}$ is used at the transmitter side, the transmission rate of the information bits is denoted as D_i^c , whereas the transmission rate D_c^c over the channel is now given by $D_c^c = \frac{n}{k} D_i^c$, since what is actually transmitted is now the coded bits rather than the information bits. The bandwidth necessary for transmission is $B_c \propto D_c^c$.

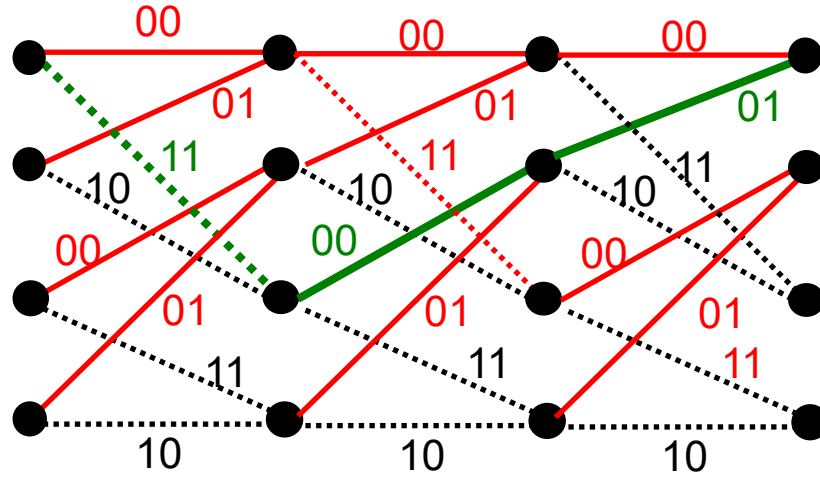
In order to keep the bandwidth unchanged when a channel code is inserted into a previously uncoded system, i.e. $B_c = B_u$, we must keep the transmission rate over the channel unchanged, i.e. $D_c^c = D_c^u$. As we have $D_c^c = \frac{n}{k} D_i^c$, the only way to have $D_c^c = D_c^u$ is to ensure that $D_i^c = \frac{k}{n} D_i^u$. In other words, the rate at which information bits are transmitted must be decreased by a factor $\frac{n}{k} > 1$.

If we wish the transmission rate of information bits to be kept unchanged when a channel code is inserted into a previously uncoded system, i.e. $D_i^c = D_i^u$, we need to have $D_c^c = \frac{n}{k} D_i^c$, which implies that $B_c = \frac{n}{k} B_u$. In other words, the bandwidth necessary for transmission must be increased by a factor $\frac{n}{k} > 1$.

Question 3



(b) We must search for the path with the smallest Hamming weight that leaves the state 00 and goes back to it.



The free distance is its Hamming weight. We find one such path (shown above in green) and it has a Hamming weight of 3. Thus, we have $d_{free} = 3$.

(c) We can use the union bound expression to answer this question:

$$P_{eb} \sim \frac{N_{d_{free}}}{2} \cdot \text{erfc} \left(\sqrt{d_{free} R_c \frac{E_b}{N_0}} \right).$$

We already know that the coding rate R_c is equal to 1/2 and the free distance d_{free} is equal to 3. We still need to determine the value of the parameter $N_{d_{free}}$.

This parameter is defined as the total Hamming weight of all messages associated with the paths at minimum Hamming distance from the all-zero path. We have identified one single path that is at distance $d_{free} = 3$ from the all-zero path. If we count the number of ones in the message corresponding to this path, we find $N_{d_{free}} = 1$.

Finally, the bit error probability at high SNR is approximated by

$$P_{eb} \sim \frac{1}{2} \cdot \text{erfc} \left(\sqrt{\frac{3}{2} \frac{E_b}{N_0}} \right),$$

and the asymptotic coding gain over uncoded BPSK is equal to $10 \log_{10} \left(\frac{3}{2} \right) \sim 1.76$ dB.

(d) No, the convolutional code studied in this question is not optimal in terms of error performance. In fact, we have seen in class that there are other codes with the same coding rate of $\frac{1}{2}$ and the same constraint length of 3 that can achieve better error performance.

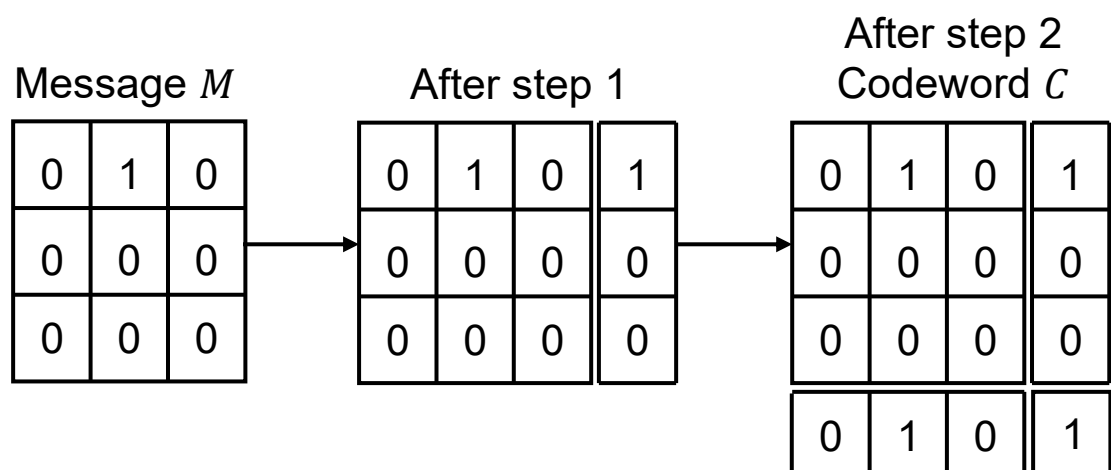
For instance, the convolutional code with generator polynomials 5 and 7 have a free distance equal to 5 and is thus able to achieve an asymptotic coding gain over uncoded BPSK of almost 4 dB. This is roughly 2.24 dB higher than the gain obtained using the code studied here.

Question 4

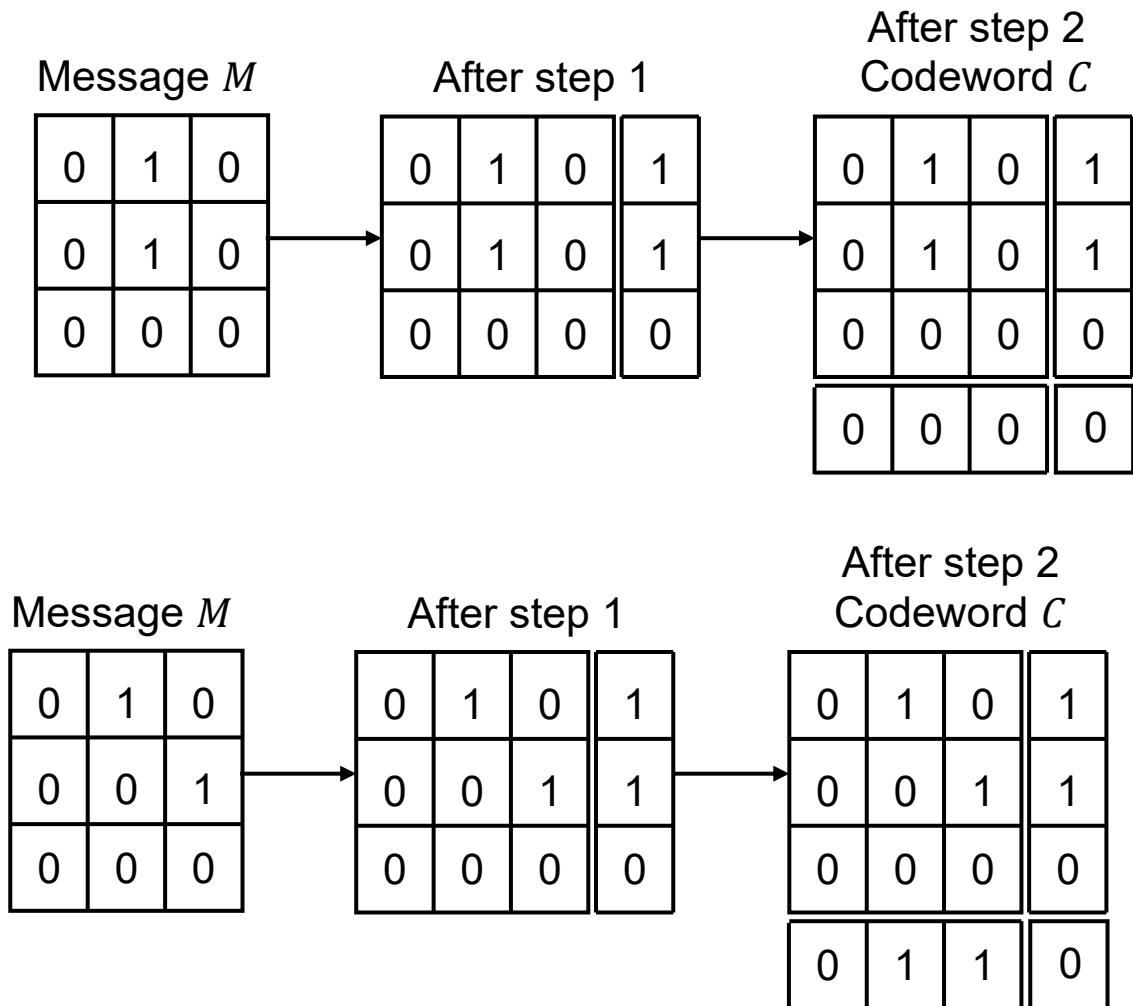
(a) ML decoding consists of computing the Hamming or Euclidean distances between the received word and all possible codewords. Then, the decoder selects the codeword which is at the minimum distance from the received word. Here, in order to implement ML decoding, the number of distances to compute is equal to 2^k . Unless k remains relatively small, there are too many computations to perform to consider ML decoding as a realistic option.

(b) We need to consider a few configurations to find the non-zero codeword with minimum Hamming weight. For instance, if we have a weight-1 message at the encoder output, the Hamming weight of the corresponding codeword is 4.

For instance, if $m = 3$:



We can see that there are no other configurations that lead to codewords with a Hamming weight smaller than 4. For instance, consider a weight-2 message with two different configurations:



We do not need to consider any other cases since these examples are sufficient to show that d_{min} cannot be smaller than 4. We thus conclude that $d_{min} = 4$ for any value of k .

(c) The asymptotic coding gain is given by $G \approx 10 \cdot \log_{10}(d_{min}R_c)$. We already know that $d_{min} = 4$. We still need to determine the expression of the coding rate R_c . This can be done by noticing that the message size is $k = m^2$, thus meaning that $m = \sqrt{k}$, whereas the codeword size is $n = (m + 1)^2 = (\sqrt{k} + 1)^2$. Finally, we obtain

$$R_c = \frac{k}{n} = \frac{k}{(\sqrt{k}+1)^2} = \left(\frac{\sqrt{k}}{\sqrt{k}+1} \right)^2,$$

and thus

$$G \approx 10 \cdot \log_{10} \left(\left(\frac{2\sqrt{k}}{\sqrt{k}+1} \right)^2 \right) = 20 \cdot \log_{10} \left(\frac{2\sqrt{k}}{\sqrt{k}+1} \right).$$

Question 5

For near-capacity operation, an error-correcting code should operate on very long blocks of information bits, i.e. $k, n \rightarrow +\infty$, and the encoding should be done in a random fashion. The main issue with such a code is the prohibitive complexity of the optimal (maximum-likelihood) decoder which requires the computation of 2^k Hamming or Euclidean distances between the received vector and all possible codewords.

Modern codes, such as turbo and low-density parity-check (LDPC) codes, possess the characteristics of near-capacity codes as their encoding is performed in a random fashion due to the use of, say, a very large random interleaver in the case of turbo codes or a very large random parity-check matrix in the case of LDPC codes. The decoding complexity issue is solved by employing an iterative algorithm of reasonable complexity which is almost as efficient as the optimal maximum-likelihood algorithm.

In contrast, more traditional codes, such as Hamming, convolutional, BCH, and Reed-Solomon codes, are designed with plenty of structure in order to minimise the decoding complexity. Their encoding is therefore not done in a random fashion and they cannot operate on very long blocks of information bits. This makes such codes unable to perform near the capacity limit.

End